

# Enterprise Infrastructure Planning for a Startup Company

## *NovaTech Startup Solutions*

ITEP 414 – System Administration and Maintenance  
Week 2 Portfolio Project

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Role: Junior System Administrator

### Part 1: Company Profile

**Company Name:** NovaTech Startup Solutions

**Nature of Business:** A software development startup that builds web and mobile apps for small businesses.

**Company Vision:** To become a trusted software solutions provider that helps small businesses grow through technology.

**Company Mission:** To build reliable, secure, and affordable software while giving our employees room to learn and grow.

**Office Location (fictional):** 3rd Floor, Meridian Business Center, Brgy. Real, Calamba City, Laguna

### Organizational Structure

NovaTech is a small startup, so the structure is pretty flat. Everyone reports to the CEO/General Manager through their department head.

CEO / General Manager

- IT Manager
- HR Manager
- Finance Manager
- Sales Manager

### Employee Distribution

| Department             | Employees |
|------------------------|-----------|
| Information Technology | 5         |
| Human Resources        | 4         |
| Finance                | 5         |
| Sales                  | 6         |
| TOTAL                  | 20        |

## Part 2: Enterprise Hardware Inventory

Below is the hardware list for all 20 employees. Desktops went to IT, HR, and Finance since they mostly work at their desks. Sales got laptops since they go out to meet clients.

| Asset ID | Hardware                | Qty | Department             | Purpose                                                             |
|----------|-------------------------|-----|------------------------|---------------------------------------------------------------------|
| HW-001   | Desktop Computer        | 5   | IT                     | For development, testing, and admin work                            |
| HW-002   | Desktop Computer        | 4   | HR                     | For records, payroll, and daily office tasks                        |
| HW-003   | Desktop Computer        | 5   | Finance                | For accounting and financial reports                                |
| HW-004   | Laptop                  | 6   | Sales                  | Sales staff go out to meet clients, so they need something portable |
| HW-005   | Server (Tower)          | 1   | IT / Shared            | Will host file sharing, Git, and internal apps                      |
| HW-006   | Router                  | 1   | Shared                 | Connects the office network to the internet                         |
| HW-007   | Switch (24-port)        | 2   | Shared                 | Connects all wired PCs, printers, and the server                    |
| HW-008   | Printer (Multifunction) | 2   | Shared                 | For printing/scanning, one near HR/Finance, one near Sales/IT       |
| HW-009   | UPS                     | 3   | Server room + 2 areas  | So the server and PCs don't shut down during brownouts              |
| HW-010   | Wireless Access Point   | 2   | Shared                 | Wi-Fi for laptops and phones                                        |
| HW-011   | NAS Storage             | 1   | IT / Shared            | Shared drive/backup storage for the whole office                    |
| HW-012   | External Backup Drive   | 2   | IT                     | Extra backup copy of important files                                |
| HW-013   | Monitor                 | 16  | IT, HR, Finance, Sales | 14 for the desktop users + 2 extra for IT                           |

## Part 3: Enterprise Software Inventory

Most of the software chosen is either free/open source or already included in the company's licenses, to keep costs down for a startup.

| Software         | Version   | License          | Purpose                                     |
|------------------|-----------|------------------|---------------------------------------------|
| Windows 11 Pro   | 22H2      | Volume License   | Main OS for all desktops/laptops            |
| Ubuntu Server    | 22.04 LTS | Free/Open Source | OS for the server (file sharing, Git, apps) |
| Microsoft Office | 365       | Subscription     | Docs, spreadsheets, and presentations       |
| VS Code          | Latest    | Free             | Code editor for the IT team                 |
| Git              | Latest    | Free/Open Source | Version control for the code                |
| GitHub Desktop   | Latest    | Free             | Easier GUI for pushing to GitHub            |
| VirtualBox       | 7.x       | Free/Open Source | For running test VMs                        |

| Software           | Version  | License             | Purpose                                  |
|--------------------|----------|---------------------|------------------------------------------|
| Google Chrome      | Latest   | Free                | Main browser for everyone                |
| Microsoft Defender | Built-in | Included w/ Windows | Basic antivirus for all PCs              |
| AnyDesk            | Latest   | Free/Business       | So IT can remote into a staff PC to help |
| 7-Zip              | Latest   | Free/Open Source    | Compressing files and backups            |

## Part 4: Enterprise Network Inventory

This is the list of networking equipment needed to set up the office LAN.

| Item            | Specification                    | Qty | Purpose                                  |
|-----------------|----------------------------------|-----|------------------------------------------|
| ISP Modem       | Fiber, up to 300 Mbps            | 1   | Connects the office to the internet      |
| Router          | Enterprise router w/ VPN support | 1   | Routes traffic in and out of the network |
| Firewall        | NGFW appliance                   | 1   | Blocks unwanted/suspicious traffic       |
| Switch          | 24-port Gigabit Managed          | 2   | Connects all wired devices               |
| Access Point    | Dual-band Wi-Fi AP               | 2   | Wireless coverage for the whole floor    |
| Patch Panel     | 24-port Cat6                     | 1   | Organizes the cable runs                 |
| CAT6 Cables     | Various lengths                  | 40  | Wiring for PCs, printers, APs            |
| RJ45 Connectors | Cat6-rated                       | 100 | For terminating the cable runs           |

## Part 5: Enterprise Network Diagram

The diagram (see attached PNG/PDF, built in Draw.io) follows this flow: Internet → ISP Modem → Router → Firewall → Core Switch. The switch then connects to the Server, Printer, Wireless AP, and the four department groups (IT, HR, Finance, Sales).

### What's in the diagram

- Internet cloud connected to the ISP Modem
- Modem to Router to Firewall (traffic gets filtered before entering the LAN)
- Firewall to the Core Switch
- Switch branches out to: Server, NAS, Printer, Wireless AP, and the 4 department groups
- Wireless AP has a dashed connection to Sales laptops (Wi-Fi)
- Title and my name are labeled on the diagram

Diagram file: diagrams/NovaTech\_Network\_Diagram.drawio — open it in draw.io, check it, then export as PNG and PDF.

## Part 6: System Administration Roles

## **Helpdesk Technician**

### **Responsibilities:**

- First point of contact when an employee has a computer problem
- Logs and tracks support tickets
- Sets up new PCs and installs software
- Escalates harder issues to network/system admins

### **Skills:**

- Good communication skills
- Basic Windows/Mac troubleshooting
- Patience
- Documentation habits

### **Common Tools:**

- Ticketing system (e.g. Zendesk)
- AnyDesk/TeamViewer
- MS Office

### **Certifications:**

- CompTIA A+
- ITIL Foundation

## **Network Administrator**

### **Responsibilities:**

- Sets up and maintains routers, switches, firewall, and Wi-Fi
- Monitors the network for problems
- Sets up security rules and VPN
- Plans network upgrades

### **Skills:**

- Understands TCP/IP, routing, switching
- Firewall configuration
- Network troubleshooting

### **Common Tools:**

- Cisco Packet Tracer
- Wireshark
- PRTG/Nagios

### **Certifications:**

- CompTIA Network+
- Cisco CCNA

## **Linux System Administrator**

### **Responsibilities:**

- Sets up and maintains the Linux server

- Manages user accounts and file permissions
- Automates tasks with scripts
- Applies updates and manages backups

**Skills:**

- Comfortable with the command line/Bash
- Knows Linux file systems and permissions
- Scripting

**Common Tools:**

- SSH
- cron
- Bash
- rsync

**Certifications:**

- LPIC-1
- RHCSA

## **Cloud Administrator**

**Responsibilities:**

- Manages cloud resources (backup, hosting, email)
- Sets up cloud accounts and access permissions
- Watches cloud costs
- Plans for future cloud migration

**Skills:**

- Knows cloud basics (IaaS/PaaS/SaaS)
- Cloud security/IAM
- Cost management

**Common Tools:**

- Azure Portal / AWS Console
- Terraform

**Certifications:**

- Azure Administrator Associate
- AWS Solutions Architect Associate

## **How they work together**

At NovaTech, these four roles depend on each other. The Helpdesk Technician handles the everyday small issues and passes on the bigger ones. If it's a connectivity or security issue, it goes to the Network Administrator. If it's about the server (files, Git, internal apps), it's the Linux Admin's job. And once the company starts using more cloud services (like cloud backup), the Cloud Administrator takes care of that side. They basically form a chain from the end user all the way to the servers and the cloud, so they need to communicate well with each other to keep things running.

## **Part 7: Infrastructure Recommendations**

### **Internet Provider**

A business-grade fiber plan (at least 300 Mbps) from a local ISP like PLDT Business or Converge Business. Business plans usually come with a static IP and faster support response, which matters since the company depends on the internet for daily work.

### **Server Specifications**

A single tower server with an 8-core CPU, 32GB RAM, and RAID storage (SSD + HDD combo) running Ubuntu Server. This should be enough for file sharing, Git, and a small internal app for 20 users, with room to grow.

### **Backup Strategy**

Follow the 3-2-1 rule: 3 copies of important data, on 2 different storage types (NAS + external drive), with 1 copy off-site/cloud. Backups should run automatically every night and get tested regularly so we know they actually work.

### **Security Recommendations**

Separate the network into VLANs per department, turn on the firewall's intrusion detection, use SSH keys instead of passwords for server access, and require VPN for anyone connecting remotely.

### **Antivirus**

Microsoft Defender is already built into Windows 11 Pro, so it makes sense to use that as the base protection, managed centrally so IT can check the status of all PCs from one place.

### **Password Policy**

At least 12 characters, mixed case + numbers + symbols, with MFA required for email, VPN, and server logins. A password manager (like Bitwarden) should be given to staff so they don't reuse passwords.

### **Expansion Plan**

The network already has a second switch and extra AP included, so adding more staff later won't need a full redesign. Once the company grows past around 40-50 people, it's probably time to start moving some services to the cloud.

## **Part 8: Personal Reflection**

Before this project, I thought being a System Administrator was mostly about fixing computers when something goes wrong. Doing this project showed me that a huge part of the job actually happens before anything gets set up — figuring out what the company actually needs and planning it out properly. Making the hardware and software lists wasn't just about listing things, I had to think about why each department needed what, like giving Sales laptops instead of desktops since they're not always in the office.

The part I found hardest was the network diagram. Listing equipment in a table was easy, but figuring out how everything should connect logically and still look clean and readable took a few tries. I also learned a lot from researching the four IT roles — I didn't realize how much these positions actually depend on and work with each other.

This project made it clear why planning matters before deployment. If a company just buys computers and internet without planning, they'll probably end up with the wrong equipment, no security, and no backups, and fixing that later is way more expensive and annoying than doing it right from the start.

Overall, this helped me practice the kind of documentation a real System Administrator would need to make and defend — inventories, diagrams, and recommendations that are actually justified, not just picked randomly. It made me more confident about approaching a new company's IT setup step by step instead of just guessing.